Amutheezan Sivagnanam

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SUMMARY

PhD Candidate and **Research Assistant** specializing in **Machine Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, ML **training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python**, **TensorFlow**, **PyTorch**, AWS, and agile CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

EDUCATION

Pennsylvania State University
Ph.D., Computer Science — *Aug 2025*

University of Houston
M.S., Computer Science — *Aug 2022*

WORK EXPERIENCE

Pennsylvania State University — Graduate Research Assistant
Applied Artificial Intelligence Lab
Aug. 2022 - May 2025

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).
- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problems to address these challenges.
- Applied artificial intelligence–based solutions to tackle real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences and presented results at DOE and NSF meetings.
- Developed a deep reinforcement learning (DRL) method for proactive responder repositioning in emergency management, achieving response times 1000× faster while reducing operational delays.
- Applied DRL to solve online vehicle routing with advance booking, enabling real-time confirmations within seconds.

The University of Houston — Graduate Research Assistant
Resilient Networks and Systems Lab
Sep. 2019 - Aug. 2022

- Spearheaded research projects funded by the U.S. Department of Energy (DOE) and the National Science Foundation (NSF).
- Studied real-world decision-making problems and identified gaps in existing solution approaches.
- Developed mathematical models and formulated problem statements to effectively address these challenges.
- Applied AI-based solution approaches to real-world problems and successfully deployed them in relevant industries.
- Published research findings in top-tier AI conferences and presented results at DOE and NSF meetings.
- Introduced heuristics that reduced annual energy costs by $140K for public transit agencies operating mixed fleets of buses.
- Implemented a deep reinforcement learning approach that reduced operational costs by 20% by enabling online booking for traditionally offline Vehicle Routing Problems (VRPs).
- Collected and analyzed real-world data using Python to identify trends—such as changes in paratransit operations before and after COVID-19 and the impact of vulnerability reward programs.

LSEG Technology — Software Engineer
Jan 2018 – Jul 2019

- Implemented unit testing for libraries in the Post-Trade C++ codebase, enhancing software reliability and ensuring adherence to industry standards.
- Validated and executed database changes for Post-Trade products for the Singapore Stock Exchange using Behavior Driven Development (BDD) in Java.
- Contributed to the CI/CD pipeline by integrating Python scripts and Git version control to streamline deployments and optimize performance.

WSO2 Lanka PVT Ltd — Software Engineering Intern
Jul 2016 – Dec 2016

- Developed an alert generation system analyzing descriptive HL7/FHIR data to monitor disease outbreaks, triggering timely email and SMS notifications.
- Engineered a mechanism to evaluate hospital functionality by assessing admission and discharge messages, including bed and oxygen cylinder availability.

SKILLS

Programming Languages: Python, C++, Java, C, JavaScript, PHP, MATLAB
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Transformers, Deep Learning, Reinforcement Learning, DQN, DDPG, TRPO, PPO, GRPO, RLHF, Bayesian Optimization, Clustering, Statistical Analysis, Data Mining, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib
Databases: SQL, MySQL, MongoDB, SQLite
Tools & Platforms: Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash
Methodologies & Practices: Object-Oriented Development, Agile Practices, Behaviour Driven Testing
Optimization & Operations: Linear Programming, Integer Programming, CPLEX, Gurobi, Mosek, OR-Tools

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PUBLICATIONS

CONFERENCE PROCEEDINGS

[C7] **Sivagnanam, A.,** Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the International Conference on Machine Learning (ICML-24)*.

[C6] R. Sen, **A. Sivagnanam,** A. Laszka, A. Mukhopadhyay and A. Dubey, "Grid-Aware Charging and Operational Optimization for Mixed-Fleet Public Transit," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 4172-4179,

[C5] Pavia, S., Rogers, D., **Sivagnanam, A**., Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 8767–8770).

[C4] Atefi, S., **Sivagnanam, A.,** Ayman, A., Grossklags, J., & Laszka, A. (2023, May). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference* (pp. 2209–2219).

[C3] **Sivagnanam, A**., Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).

[C2] **Sivagnanam, A**., Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938).*

[C1] Ayman, A., Wilbur, M., **Sivagnanam, A.,** Pugliese, P., Dubey, A., & Laszka, A. (2020, September). *Data-driven prediction of route-level energy use for mixed-vehicle transit fleets.* In Proceedings of the 6th IEEE International Conference on Smart Computing (SMARTCOMP) (pp. 1–8). IEEE.

JOURNALS

[J2] Wilbur, M., Ayman, A., **Sivagnanam, A.,** Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on Public Transit Accessibility and Ridership. *Transportation Research Record*, *2677*(4), 531-546.

[J1] Ayman, A., **Sivagnanam, A**., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2022). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology, 22*(1), Article 7, 1–29

WORKSHOPS

[W1] **Sivagnanam, A.,** Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.,** Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.),** *Artificial Intelligence and Society* (book chapter). ACM Press.

SERVICES

Program Committee Member
International Joint Conference on Artificial Intelligence 2025

Reviewer
International Joint Conference on Artificial Intelligence 2024

Reviewer
AI4Research Workshop @IJCAI2024 2024

Auxiliary Reviewer
22nd International Conference on Autonomous Agents and Multiagent Systems 2023